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Batch:**B4**

Kubernetes Apache Web App Deployment Lab

Objective

To deploy and manage an Apache web application using Kubernetes.

Steps Performed

- Created deployment using httpd image
- Exposed deployment using NodePort service
- Accessed application using port-forward
- Scaled deployment to multiple replicas
- Introduced error (wrong image)
- Diagnosed issue using kubectl describe
- Fixed deployment image
- Tested self-healing by deleting a pod

Commands Used

```
kubectl create deployment apache --image=httpd
kubectl expose deployment apache --port=80 --type=NodePort
kubectl get pods
kubectl scale deployment apache --replicas=2
kubectl set image deployment/apache httpd=wrongimage
kubectl describe pod <pod-name>
kubectl set image deployment/apache httpd=httpd
kubectl delete pod <pod-name>
```

Observations

- Pods were created successfully
- Application was accessible via port-forward

- Scaling created multiple pods
- Wrong image caused ImagePullBackOff error
- Deployment recovered after fixing the image
- Kubernetes recreated deleted pods automatically

Conclusion

Kubernetes simplifies application deployment, scaling, and management. It provides self-healing capabilities and helps handle failures efficiently.